# Math Assignment 2

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(1)

Find the matrix representation of the Linear transformation $T: \mathbb{R}^3 \to \mathbb{R}^2$ defined as $T(x, y, z) = (2x + 3y - z, 4x - y + 2z)$ with respect to the bases $B = \{(1,1,0), (1,2,3), (1,3,5)\}$ and $B' = \{(1,2), (2,3)\}$.

$T(v_1) = T(1,1,0) = (2(1)+3(1)-0, 4(1)-1+2(0)) = (5, 3)$

$T(v_1) = \begin{bmatrix} 5 \\ 3 \end{bmatrix} = x\begin{bmatrix} 1 \\ 2 \end{bmatrix} + y\begin{bmatrix} 2 \\ 3 \end{bmatrix}$

$x + 2y = 5 \qquad y = 7$

$2x + 3y = 3 \qquad x = -9$

$C_1 = \begin{bmatrix} -9 \\ 7 \end{bmatrix}$

$T(v_2) = T(1,2,3) = (2(1)+3(2)-3, 4(1)-2+2(3)) = (2+6-3, 4-2+6) = (5, 8)$

$T(v_2) = \begin{bmatrix} 5 \\ 8 \end{bmatrix} = x\begin{bmatrix} 1 \\ 2 \end{bmatrix} + y\begin{bmatrix} 2 \\ 3 \end{bmatrix}$

$x + 2y = 5 \qquad \Rightarrow x = 1$

$2x + 3y = 8 \qquad y = 2$

$C_2 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$

$T(v_3) = T(1,3,5) = (2(1)+3(3)-5, 4(1)-3+2(5)) = (6, 11)$

$T(v_3) = \begin{bmatrix} 6 \\ 11 \end{bmatrix} = x\begin{bmatrix} 1 \\ 2 \end{bmatrix} + y\begin{bmatrix} 2 \\ 3 \end{bmatrix}$

$x + 2y = 6 \qquad y = 1$

$2x + 3y = 11 \qquad x = 4$

$C_3 = \begin{bmatrix} 4 \\ 1 \end{bmatrix}$

$$[T]_B^{B'} = \begin{bmatrix} -9 & 1 & 4 \\ 7 & 2 & 1 \end{bmatrix}$$

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Consider the bases $S = \{(1,2), (2,3)\}$ and $S' = \{(1,3), (1,4)\}$ of $\mathbb{R}^2$. Find the change of basis matrix: (a) $P$ from $S$ to $S'$ (b) $Q$ from $S'$ to $S$.

$S = \{(1,2), (2,3)\}$ $\quad$ $S' = \{(1,3), (1,4)\}$

$S_1 = (1,2)$
$S_2 = (2,3)$

$S_1' = (1,3)$
$S_2' = (1,4)$

a) $P_{S \to S'}$

$(1,2) = c_1(1,3) + c_2(1,4)$

$c_1 + c_2 = 1$
$3c_1 + 4c_2 = 2$
$\Rightarrow c_1 = 2$
$c_2 = -1$
$P_1 = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$

$(2,3) = d_1(1,3) + d_2(1,4)$

$d_1 + d_2 = 2$
$3d_1 + 4d_2 = -3$
$\Rightarrow d_1 = 5$
$d_2 = -3$
$P_2 = \begin{bmatrix} 5 \\ -3 \end{bmatrix}$

$$P_{S \to S'} = \begin{bmatrix} 2 & 5 \\ -1 & -3 \end{bmatrix}$$

b) $Q_{S' \to S} = (P_{S \to S'})^{-1}$

$$Q_{S' \to S} = \begin{bmatrix} 2 & 5 \\ -1 & -3 \end{bmatrix}^{-1} \Rightarrow \frac{1}{-1} \begin{bmatrix} -3 & -5 \\ 1 & 2 \end{bmatrix}$$

$$Q_{S' \to S} = \begin{bmatrix} 3 & 5 \\ -1 & -2 \end{bmatrix}$$

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Find the angle $\theta$ between $u = \begin{bmatrix} 2 & 1 \\ 3 & -1 \end{bmatrix}$ and $v = \begin{bmatrix} 0 & -1 \\ 2 & 3 \end{bmatrix}$.

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$$U = \begin{bmatrix} 2 & 1 \\ 3 & -1 \end{bmatrix} \quad V = \begin{bmatrix} 0 & -1 \\ 2 & 3 \end{bmatrix}$$

$$\cos\theta = \frac{\langle U, V \rangle}{\|U\| \|V\|}$$

$$\langle U, V \rangle = (2)(0) + (1)(-1) + (3)(2) + (3)(-1) = 0 - 1 + 6 - 3 = 2$$

$$\|U\|^2 = \langle U, U \rangle = 2^2 + 1^2 + 3^2 + (-1)^2 \Rightarrow \|U\|^2 = 15$$
$$\|U\| = \sqrt{15}$$

$$\|V\|^2 = \langle V, V \rangle = 0^2 + (-1)^2 + 2^2 + 3^2 \Rightarrow \|V\|^2 = 14$$
$$\|V\| = \sqrt{14}$$

$$\cos\theta = \frac{\langle U, V \rangle}{\|U\| \|V\|}$$

$$= \frac{2}{\sqrt{15}\sqrt{14}} = \frac{2}{\sqrt{210}}$$

$$\theta = \cos^{-1}\left(\frac{2}{\sqrt{210}}\right)$$

$$\theta = 82.067^\circ$$